

Quantized Markov chain couplings that prepare Qsamples

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January 14, 2026

Based on

Quantized Markov chain couplings that prepare Q samples

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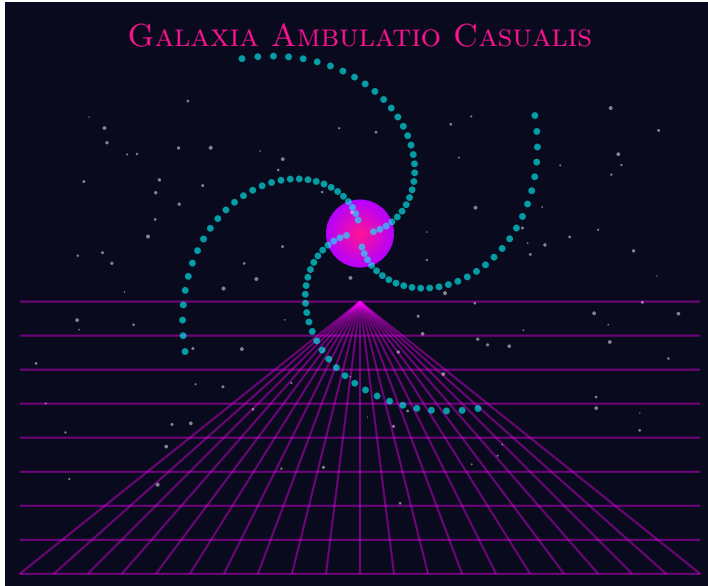
Quantum 9, 1951 (2025), <https://arxiv.org/abs/2504.02651>

Quantum algorithms

With steady progress in quantum hardware, error correction, and fault tolerance,

the search for new quantum algorithms for large-scale fault-tolerant quantum computers has become more critical than ever.

Galactic Scale Research Description



Sampling from probability distributions

- ▶ Many important problems in computer science, physics, and machine learning can be solved by sampling from interesting probability distributions.
- ▶ Markov chains are an important algorithmic tool for efficiently generating such samples.
- ▶ For instance, one application is approximating the best route for the traveling salesman problem using simulated annealing, which employs a Markov chain parameterized by temperature.
- ▶ Well-designed Markov chains enable us to cleverly explore exponentially large state spaces.

Qsamples – quantum encodings of probability distributions

- ▶ Let $\pi = (\pi_x : x \in \Omega)$ be a probability distribution over a finite state space Ω .
- ▶ The qsample $|\sqrt{\pi}\rangle$ is a quantum state encoding the classical state π .
- ▶ It is given by

$$|\sqrt{\pi}\rangle = \sum_{x \in \Omega} \sqrt{\pi_x} |x\rangle.$$

- ▶ Quantum algorithms leverage qsamples to achieve speed-ups.

Examples of applications of qsamples I

- ▶ Preparing a uniform superposition $|G\rangle$ over all group elements of a finite group G

$$|G\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle$$

is the first step in quantum algorithms for solving the hidden subgroup problem (e.g., Shor's DLOG quantum algorithm).

Examples of applications of qsamples II

- ▶ Decoded Quantum Interferometry prepares a qsample – a purification – of the density matrix

$$\sigma = \frac{\mathcal{P}^2(H)}{\text{Tr}(\mathcal{P}^2(H))}$$

for a Hamiltonian H and a polynomial $\mathcal{P}$.

- ▶ The qsample has the form

$$(\sqrt{\sigma} \otimes I)|\Phi\rangle$$

where $|\Phi\rangle$ is the (unnormalized) maximally entangled state.

Examples of applications of qsamples III

- ▶ A quantum sample $|\sqrt{\pi}\rangle$ is a crucial ingredient for the quantum Chebyshev method, which enables one to reduce the cost from $O(1/\varepsilon^2)$ to $O(1/\varepsilon)$ when estimating the mean of a random variable with relative error ε .
- ▶ A quantum sample $|\sqrt{\pi}\rangle$ is key to obtaining speed-ups in other applications, such as finding marked elements, testing properties of distributions etc.

Comments on the hardness of preparing qsamples

- ▶ Preparing qsamples of general probability distributions is very hard.
- ▶ Problems in SZK (Statistical Zero Knowledge) could be solved on quantum computers with the help of qsamples of probability distributions that are generated by classical circuits.
- ▶ Preparing qsamples of stationary distribution for general Markov chains is very difficult.
- ▶ For instance, one could solve graph isomorphism with the help of the uniform superposition of all graphs isomorphic to a given graph.

Markov chain terminology I

- ▶ Let $P = (p_{x',x})$ denote the transition matrix of an ergodic Markov chain on state space Ω with stationary distribution $\pi = (\pi_x)$.
- ▶ For any initial probability distribution q , we have

$$\lim_{m \rightarrow \infty} P^m q = \pi.$$

Markov chain terminology II

- Define

$$d(m) = \max_{x \in \Omega} \frac{1}{2} \sum_{x' \in \Omega} |[P^m]_{x',x} - \pi_{x'}|,$$

where

$$([P^m]_{x',x} : x' \in \Omega)$$

is the probability distribution that we obtain by starting the Markov chain in the initial state x and making m transitions.

- The total variation distance $d(m)$ measures how close we are to the desired stationary distribution π as the number of steps m increases.

Markov chain terminology III

- The mixing time $t_{\text{mix}}(\varepsilon)$ is defined by

$$t_{\text{mix}}(\varepsilon) = \min\{m : d(m) \leq \varepsilon\}$$

for $\varepsilon \in (0, 1)$.

- It satisfies the bound

$$t_{\text{mix}}(\varepsilon) \leq \lceil \log_2 \varepsilon^{-1} \rceil t_{\text{mix}},$$

where $t_{\text{mix}} = t_{\text{mix}}(1/4)$.

Galactic Scale Research Summary I

- Given a certain Markov chain P with stationary distribution π , we construct an efficiently implementable quantum channel $\mathcal{T}$ such that

$$\lim_{m \rightarrow \infty} \mathcal{T}^m(\rho) = |\sqrt{\pi}\rangle\langle\sqrt{\pi}|.$$

for all initial states ρ .

Galactic Scale Research Summary II

- We bound the convergence, i.e., how fast the trace norm distance

$$\frac{1}{2} \|\mathcal{T}^m(\rho) - |\sqrt{\pi}\rangle\langle\sqrt{\pi}|\|_{\text{tr}}$$

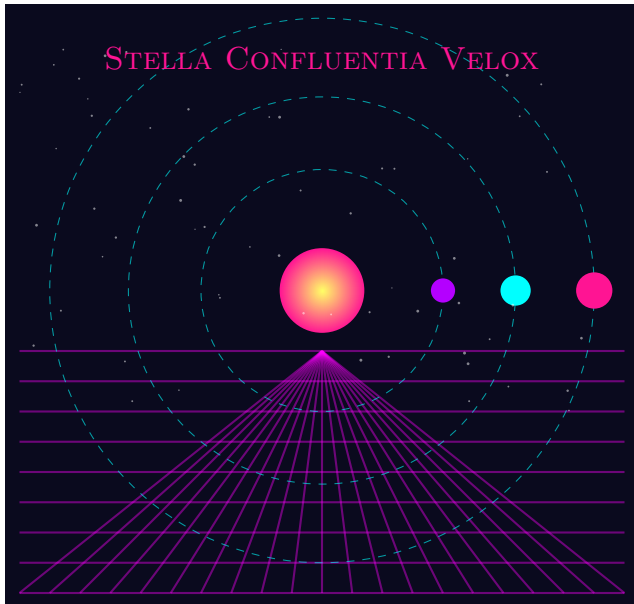
decays as m increases, using certain parameters of the Markov chain P .

- The trace norm $\|\bullet\|_{\text{tr}}$ is defined by

$$\|A - B\|_{\text{tr}} = \text{tr}\left(\sqrt{(A - B)(A - B)}\right)$$

for hermitian matrices A and B .

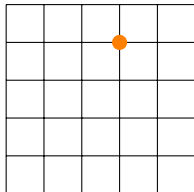
Stellar Scale Research Description



Markov chain coupling

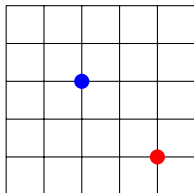
- ▶ Markov chain coupling is one of the most important techniques for proving fast mixing.
- ▶ Fast mixing means that t_{mix} scales poly-logarithmically in $|\Omega|$.
- ▶ Markov chain couplings are at the heart of our novel quantization method.

Drunk Eve in Manhattan



- ▶ Drunk Eve moves randomly around Manhattan.
- ▶ At each time step, Eve measures two $|+\rangle$ states in the computational basis.
- ▶ The two bits determine whether Eve moves up, down, left, or right, but she cannot leave the grid.
- ▶ Eve stays where she is if no move is possible.
- ▶ The stationary distribution π is approximately uniform (the effects due to the boundaries diminish as the size increases).

Drunk Alice and Bob in Manhattan



- ▶ Drunk Alice and Bob move randomly around Manhattan.
- ▶ At each time step, Alice and Bob measure the halves of two Bell states $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ shared between them.
- ▶ The rules for moving are the same as in the Eve example.
- ▶ Viewed individually, they move like Eve.
- ▶ It's also easy to see that eventually Alice and Bob will find each other.
- ▶ Once Alice and Bob meet, they stay together forever.

Formal definition of Markov chain coupling

A coupling is a Markov chain on state space $\Omega \times \Omega$ with transition matrix $C = (c_{(x',y'),(x,y)})$ so that:

1. Each component viewed separately undergoes transitions according to the original transition matrix P :

$$\sum_{y'} c_{(x',y'),(x,y)} = p_{x',x}$$

$$\sum_{x'} c_{(x',y'),(x,y)} = p_{y',y}$$

2. The two components remain together forever once they have met, that is, $x' = y'$ if $x = y$:

$$c_{(x',x'),(x,x)} = p_{x',x}$$

$$c_{(x',y'),(x,x)} = 0 \text{ for } x' \neq y'$$

3. The two components satisfy the symmetry condition:

$$c_{(x',y'),(x,y)} = c_{(y',x'),(y,x)}$$

Connection between coalescence and mixing times I

- ▶ Let (X_m, Y_m) denote the random state of the coupling that starts in the initial state (x, y) and makes m transitions according to the transition matrix C .
- ▶ Let τ_{coal} denote the random variable

$$\tau_{\text{coal}} = \min\{m : X_\ell = Y_\ell \text{ for all } \ell \geq m\},$$

which is called the coalescence time for the initial state (x, y) .

Connection between coalescence and mixing times II

- ▶ The total variation distance $d(m)$ and the mixing time t_{mix} are bounded from above by

$$d(m) \leq \max_{x,y} \Pr(\tau_{\text{coal}} > m)$$

$$t_{\text{mix}} \leq 4 \max_{x,y} \mathbb{E}_{x,y}(\tau_{\text{coal}}).$$

- ▶ Thus, the problem of showing that the Markov chain with transition matrix P mixes fast reduces to the problem of constructing a suitable coupling with transition matrix C that coalesces quickly when starting in any initial state (x, y) .
- ▶ See *Markov Chains and Mixing Times* by David A. Levin and Yuval Peres, 2nd edition.

Coupling time

- ▶ In analogy to the mixing time t_{mix} , it is convenient for our analysis to introduce the coupling time t_{couple} .
- ▶ We define it as

$$t_{\text{couple}} = \min \left\{ m : \max_{x,y} \Pr(\tau_{\text{coal}} > m) \leq \frac{1}{4} \right\}.$$

Stellar Scale Research Intuition

- ▶ Using quantum notation, we can express the transition matrix $C = (c_{(x',y'),(x,y)}) \in \mathcal{L}(\mathbb{C}^N \otimes \mathbb{C}^N)$ as

$$C = \sum_{x',y',x,y \in \Omega} c_{(x',y'),(x,y)} (|x'\rangle \otimes |y'\rangle)(\langle x| \otimes \langle y|).$$

- ▶ The classical map C acts on ket vectors of the form $|x\rangle \otimes |y\rangle$ and maps them to convex combinations of ket vectors $|x'\rangle \otimes |y'\rangle$.
- ▶ In contrast, a quantum map $\mathcal{E} \in \mathcal{L}(\mathbb{C}^{N \times N})$ must act on outer products of the form $|x\rangle\langle y|$ and map them to linear combinations of outer products $|x'\rangle\langle y'|$.
- ▶ Question: How can we cook up a useful quantum map $\mathcal{T}$ based on the transition matrix C ?

Stellar Scale Research Summary I

- ▶ Let $C = (c_{(x',y'),(x,y)})$ be the transition matrix of an arbitrary coupling of an ergodic Markov chain with transition matrix $P = (p_{x',x})$ and stationary distribution $\pi = (\pi_x)$.
- ▶ Define

$$\mathcal{T}^*(\bullet) = D^{-1/2} \mathcal{C}^*(D^{1/2} \bullet D^{1/2}) D^{-1/2} \quad \text{where}$$

$$\mathcal{C}^*(\bullet) = \sum_{x,y,x',y' \in \Omega} c_{(x',y'),(x,y)} |x'\rangle \langle x| \bullet |y\rangle \langle y'|$$
$$D = \text{diag}(\pi_x : x \in \Omega).$$

- ▶ The quantized coupling $\mathcal{T}$ is defined as the dual of the linear map $\mathcal{T}^*$ with respect to the Hilbert-Schmidt inner product.

Stellar Scale Research Summary II

- ▶ Let C be any coupling.
- ▶ The quantized coupling $\mathcal{T}$ is trace preserving.
- ▶ The quantized coupling $\mathcal{T}$ has $|\sqrt{\pi}\rangle\langle\sqrt{\pi}|$ as fixed point.

Stellar Scale Research Summary III

- ▶ Assume that C satisfies certain additional conditions so that the corresponding $\mathcal{T}$ is also completely positive.
- ▶ Then, given any initial state ρ , the quantized coupling $\mathcal{T}$ converges to the qsample $|\sqrt{\pi}\rangle\langle\sqrt{\pi}|$ in trace norm distance as

$$\frac{1}{2} \|\mathcal{T}^m(\rho) - |\sqrt{\pi}\rangle\langle\sqrt{\pi}|\|_{\text{tr}} \leq \sqrt{\varepsilon}$$

after

$$m \geq \frac{1}{2} \log_2 \left(\frac{1}{\varepsilon \pi_*} \right) t_{\text{couple}}$$

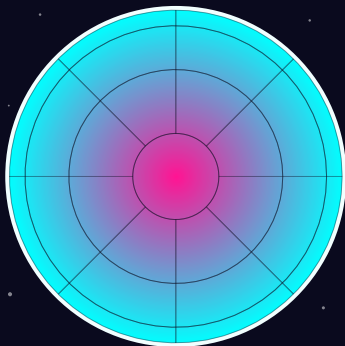
steps, where $\pi_* = \min\{\pi_x : x \in \Omega\}$.

Stellar Scale Research Summary IV

- ▶ For independent and grand couplings, the resulting quantized coupling $\mathcal{T}$ is completely positive.
- ▶ The quantized coupling $\mathcal{T}$ can be implemented efficiently for sparse grand couplings.

Planetary Scale Research Description

PLANETA CREATIO EXPEDITA



Proof of fixed point I

- The qsample $|\sqrt{\pi}\rangle$ is a fixed point of the quantized coupling $\mathcal{T}$, i.e.,

$$\mathcal{T}(|\sqrt{\pi}\rangle\langle\sqrt{\pi}|) = |\sqrt{\pi}\rangle\langle\sqrt{\pi}|.$$

Proof of fixed point II

- Recall that the matrix D is given by

$$D = \text{diag}(\pi_x : x \in \Omega).$$

.

- We write the qsamples $|\sqrt{\pi}\rangle$ as

$$\begin{aligned} |\sqrt{\pi}\rangle &= \sum_{x \in \Omega} \sqrt{\pi_x} |x\rangle \\ &= D^{1/2} |e\rangle \end{aligned}$$

where

$$|e\rangle = \sum_{x \in \Omega} |x\rangle = (1, 1, \dots, 1)^T$$

is the all-ones vector.

Proof of fixed point III

► We have

$$\begin{aligned}\mathcal{C}(|e\rangle\langle e|) &= \sum_{x,y,x',y' \in \Omega} c_{(x',y'),(x,y)} |y\rangle\langle y'|e\rangle\langle e|x'\rangle\langle x| \\ &= \sum_{x,y,x',y' \in \Omega} c_{(x',y'),(x,y)} |y\rangle\langle x| \\ &= \sum_{y,x} |y\rangle\langle x| \\ &= |e\rangle\langle e|,\end{aligned}$$

since $C = (c_{(x',y'),(x,y)})$ is column stochastic.

Proof of fixed point IV

- Inserting $|\sqrt{\pi}\rangle\langle\sqrt{\pi}|$ into

$$\mathcal{T}(\bullet) = D^{1/2} \mathcal{C}(D^{-1/2} \bullet D^{-1/2}) D^{1/2}$$

yields

$$\begin{aligned}\mathcal{T}(|\sqrt{\pi}\rangle\langle\sqrt{\pi}|) &= D^{1/2} \mathcal{C}(D^{-1/2} |\sqrt{\pi}\rangle\langle\sqrt{\pi}| D^{-1/2}) D^{1/2} \\ &= D^{1/2} \mathcal{C}(|e\rangle\langle e|) D^{1/2} \\ &= D^{1/2} |e\rangle\langle e| D^{1/2} \\ &= |\sqrt{\pi}\rangle\langle\sqrt{\pi}|.\end{aligned}$$

Sketch of convergence proof I

- ▶ Let $Q = |\sqrt{\pi}\rangle\langle\sqrt{\pi}|$ and $Q^\perp = I - Q$.
- ▶ Assume we can find an integer m for some $\varepsilon > 0$ so that

$$\mathrm{tr}[Q^\perp \mathcal{T}^m(\rho)] \leq \varepsilon$$

for all initial states ρ .

- ▶ Then, we have

$$\|\mathcal{T}^k(\rho) - |\sqrt{\pi}\rangle\langle\sqrt{\pi}|\|_{\mathrm{tr}} \leq 2\sqrt{\varepsilon}$$

for all $k \geq m$.

- ▶ The proof of this technical result relies on some fundamental tools in quantum information processing, such as the gentle measurement lemma and the data processing inequality.

Sketch of convergence proof II

► We have

$$\begin{aligned}\mathrm{tr}[Q^\perp \mathcal{T}^m(\rho)] &= \mathrm{tr}[(\mathcal{T}^*)^m(Q^\perp) \rho] \\ &\leq \mathrm{tr}[(\mathcal{T}^*)^m(Q^\perp)] \\ &= \mathrm{tr}[D^{-1/2}(C^*)^m(D^{1/2}Q^\perp D^{1/2})D^{-1/2}] \\ &\leq \frac{1}{\pi_*} \mathrm{tr}[(C^*)^m(D^{1/2}Q^\perp D^{1/2})]\end{aligned}$$

where $\pi_* = \min\{\pi_x : x \in \Omega\}$.

Sketch of convergence proof III

- ▶ Moreover, we show that
 - ▶ the rescaled projector $D^{1/2}Q^\perp D^{1/2}$ can be written as a convex combination of “quasi-particles” corresponding to elementary graph Laplacians

$$\rho_{x,y} = |-_{x,y}\rangle\langle -_{x,y}|$$

where $|-_{x,y}\rangle = \frac{1}{\sqrt{2}}(|x\rangle - |y\rangle)$.

- ▶ the map $\mathcal{C}^*$ scatters $\rho_{x,y}$ into sub-stochastic combinations according to

$$\mathcal{C}^*(\rho_{x,y}) = \sum_{x',y'} c_{(x',y'),(x,y)} \rho_{x',y'}.$$

- ▶ the power $(\mathcal{C}^*)^m$ annihilates the quasi-particle $\rho_{x,y}$ whenever $\mathcal{C}^m$ coalesces (x,y) .

Grand couplings

- ▶ A random mapping representation of the transition matrix $P = (p_{x',x})$ on state space Ω is a function $f : \Omega \times \mathcal{R} \rightarrow \Omega$, along with a $\mathcal{R}$ -valued random variable R , satisfying

$$\Pr(f(x, R) = x') = p_{x',x}$$

for all possible pairs of current state x and successor state x' .

- ▶ The transition matrix P can be expressed as

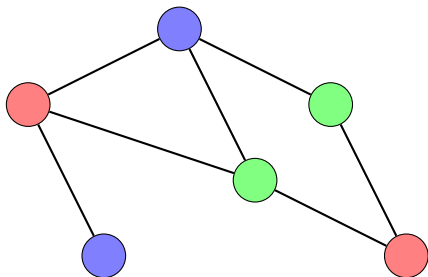
$$P = \sum_{r \in \mathcal{R}} \sum_{x \in \Omega} \Pr(r) |f(x, r)\rangle \langle x|.$$

- ▶ The transition matrix C_{grand} of the corresponding grand coupling can be expressed as

$$C_{\text{grand}} = \sum_{r \in \mathcal{R}} \sum_{x, y \in \Omega} \Pr(r) (|f(x, r)\rangle \otimes |f(y, r)\rangle) (\langle x| \otimes \langle y|)$$

Example of grand coupling – random colorings I

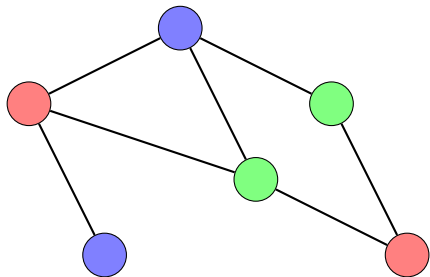
- ▶ Let $G = (V, E)$ be a graph with vertex set V and edge set E .
- ▶ A proper q -coloring of G is an element x of $\{1, \dots, q\}^V$, the set of functions from V to $\{1, \dots, q\}$, such that $x(v) \neq x(w)$ for all edges $\{v, w\} \in E$.



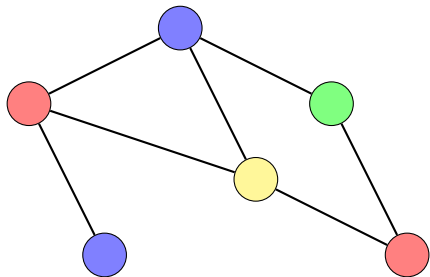
Example of grand coupling – random colorings II

- ▶ Let $\mathcal{X}$ denote the collection of all proper q -colorings of G and π denote the uniform distribution of on $\mathcal{X}$.
- ▶ A Metropolis chain for π can be constructed as follows:
 - ▶ select a vertex $v \in V$ uniformly and then select a color $k \in \{1, \dots, q\}$ uniformly.
 - ▶ if k is allowable at v , then vertex v is assigned the color k ; otherwise, no transition is made.

Example of grand coupling – random colorings III



select v to be the center node, and k to be yellow



Example of grand coupling – random colorings IV

- ▶ A grand coupling also generates such (v, k) uniformly and applies the corresponding recoloring update to both components.
- ▶ The bound

$$\Pr_{x,y}\{\tau_{\text{coal}} > m\} \leq n \exp\left(-m c_{\text{met}}(\Delta, q)/n\right),$$

holds for all pairs of initial q -colorings x and y .

Δ denotes the maximum degree and $c_{\text{met}} = 1 - (3\Delta)/q$.

Implementation I

- ▶ Assume that we have a random mapping representation such that
 - ▶ we can efficiently compute the images and pre-images of the function $f : \Omega \times \mathcal{R} \rightarrow \Omega$
 - ▶ we can efficiently compute the probabilities $\Pr(r)$
 - ▶ we can efficiently compute the ratios $\pi(x)/\pi(y)$ for the stationary distribution.
- ▶ Then, the Kraus operators

$$A_r = \sqrt{\Pr(r)} \sum_{x \in \Omega} D^{1/2} |x\rangle \langle f(x, r)| D^{-1/2}.$$

of the quantized coupling $\mathcal{T}$ are $|\mathcal{R}|$ -sparse.

Implementation II

- We can efficiently construct block-encodings of the sparse matrices A_r , i.e., unitaries U_r of the form

$$U_r = \begin{pmatrix} A_r & C_r \\ B_r & D_r \end{pmatrix} \\ = |0\rangle\langle 0| \otimes A_r + |0\rangle\langle 1| \otimes B_r + |1\rangle\langle 0| \otimes C_r + |1\rangle\langle 1| \otimes D_r.$$

- We can also efficiently implement the controlled unitary

$$W = \sum_{r \in \mathcal{R}} |r\rangle\langle r| \otimes U_r.$$

Implementation III

- ▶ Given any initial state $|\xi\rangle$, we can use W to prepare

$$\frac{1}{\sqrt{|\mathcal{R}|}} \sum_{r \in \mathcal{R}} |r\rangle \otimes (|0\rangle \otimes A_r |\xi\rangle + |1\rangle \otimes B_r |\xi\rangle)$$

- ▶ We now leverage oblivious amplitude amplification to prepare

$$\sum_{r \in \mathcal{R}} |r\rangle \otimes |0\rangle \otimes A_r |\xi\rangle$$

- ▶ Tracing out the first two registers yields

$$\mathcal{T}(|\xi\rangle\langle\xi|) = \sum_{r \in \mathcal{R}} A_r |\xi\rangle\langle\xi| A_r^\dagger$$

- ▶ This shows that $\mathcal{T}$ can be implemented efficiently since the above arguments hold for arbitrary $|\xi\rangle$.

Some open problems

- Improve analysis of convergence.

For instance, can we avoid the factor $1/\pi_*$?

- Generalize to the quantum setting.

Given a fast mixing quantum channel $\mathcal{E}$ with fixed point σ , can we design a quantum channel $\mathcal{E}_{\text{pure}}$ that

- (i) has the purification

$$(\sqrt{\sigma} \otimes I)|\Phi\rangle$$

as fixed point and

- (ii) converges to it quickly?

Thank you for your attention!